Enhancing agent-based electricity purchase forecasting accuracy comprehensively supports power supply and electricity spot trading

Rui Zhang
School of Information and Control Engineering, Shenyang Jianzhu University, Shenyang, Liaoning, China
zhangrui0325@163.com

Yujie Tang*
State Grid Dalian Pulandian District Power, Supply Company, Pulandian, Dalian, Liaoning, China
*Corresponding author: 1834347886@qq.com

Yanning Wang
School of Information and Control Engineering, Shenyang Jianzhu University, Shenyang, Liaoning, China
2385163617@qq.com

***Abstract*—Guided by the State Grid Corporation's 'Guidelines for Excellent Performance Management and Evaluation,' we apply the principles of excellent performance evaluation and lean management. Our main focus is on analyzing and forecasting the company's electricity sales market and power supply-demand balance, with the aim of enhancing the overall accuracy of the company's power purchase agency.On the basis of the LSTM model, this article adopts a method called Adaptive Noise Complete Set Empirical Mode Decomposition (CEEMDAN) to decompose the original power sequence into multiple intrinsic mode components and residuals. An improved CNN layer is added to further extract data features, which are then input into the improved LSTM model for prediction.**

By the end of December 2023, it is anticipated that the annual forecast error rate for residential agricultural power supply will be no higher than 4.8%, the forecast error rate for power purchase agency will be no higher than 5.1%, and the comprehensive error rate will be no higher than 5%, ranking among the top in the province.

Keywords-power purchase agency, electricity demand forecasting, power supply reliability, electricity spot trading.

I. INTRODUCTION

To comprehensively implement the spirit of national reform, aligning with the overall goals of achieving carbon peak and carbon neutrality, and adhering to the principle of 'tightening the middle and relaxing at both ends,' in the face of significant changes in the current electricity supply structure and supply-demand situation, it is essential to further advance the market-oriented reform of on-grid electricity pricing for coal-fired power generation. Additionally, the establishment of a mechanism for grid companies to act as agents for power purchase is critical. Real-time forecasting of load-side agent power purchase quantities should be performed. This is of paramount significance in achieving the orderly and smooth entry of all industrial and commercial users into the electricity market and in accelerating the development of the electricity market[1].

In line with market-oriented reform requirements, residential and agricultural electricity is guaranteed by grid enterprises and follows the existing catalog sales electricity price policy. Priority is given to sourcing from low-cost power suppliers, with any shortfalls being procured by grid enterprises through market-based means. Accurate forecasts of the next month's power purchase agency quantity are made to participate in electricity market transactions in conjunction with market conditions, allowing for the purchase of the required electricity quantity. This approach effectively ensures safe and reliable power supply for residential, agricultural, and power purchase agency business users[2], thereby achieving power supply reliability[3].

II. RESEARCH ON ELECTRICITY QUANTITY PREDICTION METHODS

A. Data preprocessing.

This article selects the daily residential electricity consumption of a certain city in China for the past three years and three months as experimental data, with a total of 1185 pieces of data. Using a total of 1005 pieces of data from January 1, 2021 to September 30, 2023 as the training set, and the data from December 31, 2023 to March 31, 2024 as the testing set.[4].

In order to accelerate model training and improve prediction accuracy, the dataset is normalized according to equation (1), and the range of processed values is [-1,1].

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

B. Adaptive noise complete set empirical mode decomposition.

This article first adopts a method called Adaptive Noise Complete Set Empirical Mode Decomposition (CEEMDAN), which is developed from Empirical Mode Decomposition (EMD) and Ensemble Empirical Mode Decomposition (EEMD). Based on EMD, it adaptively decomposes the original power sequence into multiple Intrinsic Mode Functions (IMF) and a residual (Res) with different frequencies and scales. Compared to EMD and EEMD, the CEEMDAN algorithm adds white noise during the decomposition process, which is the IMF components of each order obtained through EMD decomposition, solving the modal mixing defect in EMD and improving the residual noise problem in EEMD reconstructed signals. This method is based on empirical mode decomposition, which decomposes the signal into multiple intrinsic mode functions and uses adaptive noise set technology to remove noise from the signal and extract the essential features of the signal. Compared with traditional methods, fully adaptive noise set empirical mode decomposition has better robustness and higher signal-to-noise ratio, making it suitable for processing various complex signals. Decompose the original power sequence into multiple intrinsic mode components and residuals, add Gaussian white noise to the original signal, and obtain a new signal. The calculation formula is shown in equation (2).

$$x_i(t) = x(t) + \varepsilon a_i(t) \tag{2}$$

C. Improved Convolutional Neural Network

As a special type of feed forward neural network, CNN has the characteristics of convolutional computation and deep structure, and is one of the representative models in deep learning technology. It is widely used in time series prediction such as image recognition, image processing, electricity prediction, photovoltaic power prediction, etc. To further reduce the fluctuation of prediction errors and reduce the prediction time of the model, two one-dimensional convolutional layers are used in this CNN model to extract local features of the data. The one-dimensional convolution can use a 1xn convolution kernel to associate the information inside each output hidden layer state , reduce data noise and instability factors, improve data quality, and achieve local optimal sparsity; The pooling methods can be divided into maximum pooling, average pooling, etc. This article adopts the maximum pooling method; Finally, a fully connected layer is used to learn the data features and predict the electricity consumption of each decomposed component.The schematic diagram of CNN network structure is shown in Figure 1.

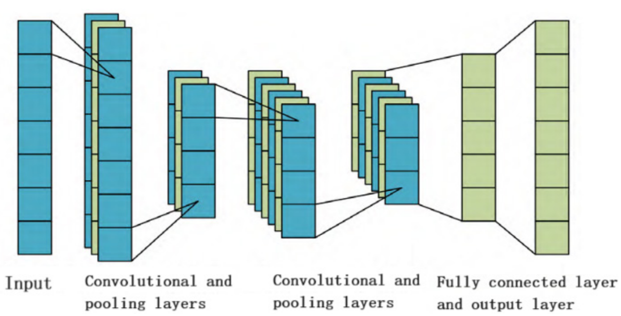

Fig.1 Schematic diagram of CNN network structure.

The convolution calculation method is shown in formula (3).

$$y(i) = f_x\left[y(i-1) \otimes W_i + b_c\right] \tag{3}$$

At present, CNN networks suffer from activation functions leading to gradient vanishing and mean shift, which hinders the further improvement of their fault diagnosis accuracy; In addition, CNN focuses on the spatial characteristics of data, and its feature extraction performance for time series is not ideal. The function of the activation function is to add nonlinear factors to the neural network. After being mapped by the activation function, the output becomes a nonlinear function, making the network more expressive.

The common activation functions include sigmoid function and tanh function. These two activation functions have many drawbacks, such as high computational complexity, easy occurrence of gradient vanishing and over fitting. Therefore, neural networks often use rectified linear units (ReLu) as the activation function. However, ReLU function has the following problems: while solving the problem of gradient vanishing, there is a mean shift. When the independent variable is in a non positive range, the neuron is in a non active state, which can easily cause the problem of subsequent network non convergence. Exponential Linear Units (ELU), a function that combines sigmoid and ReLU functions, are characterized by soft saturation on the left and no saturated rows on the right. The soft saturation part on the left has better adaptability to input changes and noise, and can solve the problem of mean shift. Therefore, in this paper, the CNN layer chooses ELU as the activation function.

D. Improved Long Short Memory Neural Network

Long Short Term Memory (LSTM) neural network is an improvement of Recurrent Neural Network (RNN), which avoids the problem of long-term dependencies and solves the problem of gradient vanishing or exploding when RNN processes long time series. By using its own three gate structure (output gate, forget gate, and output gate) and two basic units (memory cell state and hidden unit), it comprehensively analyzes the temporal characteristics of the data, completes the screening and retention of information, and better captures the dependency relationships in time series data. LSTM network is used to separately mine the temporal features of each component and obtain multiple prediction results. The forget gate decides to discard the proportion of the output information and cell state information of the previous unit in LSTM, the input gate is used to analyze the data information at the current time, and the output gate is used to output the hidden layer information of the current unit.The basic unit structure of LSTM is shown in Figure 2

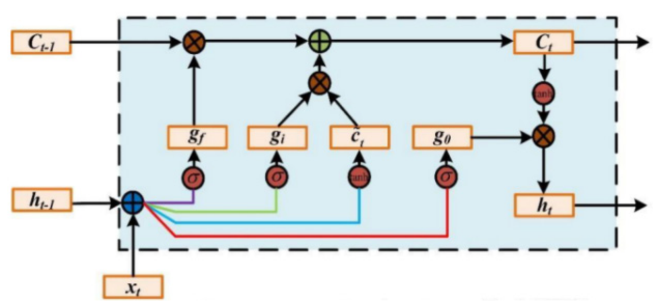

Fig.2 Improved LSTM basic unit structure.

At present, dealing with certain very long or continuous time series is still an unsolvable problem, and these time series are not pre subdivided into appropriate training subsequences and clear start and end. This study introduces peephole connections, which can enable the current cell state to capture the forgotten information from the previous time series, better capturing long-term dependencies, and compensating for the potential loss of some information by memory cells when outputting at time t+1 of the output gate. The unit structure of the improved LSTM is shown in Figure 3.

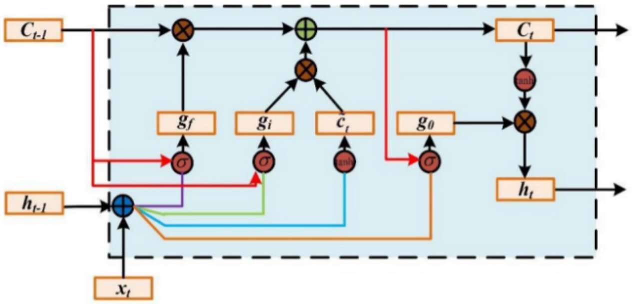

Fig.3 Improved LSTM basic unit structure.

The input gate is shown in equation (4).

$$i_t = \sigma(W_i \cdot [C_{t-1}, h_{t-1}, x_t] + b_i) \tag{4}$$

The forgetting gate is shown in equation (5).

$$f_t = \sigma(W_f \cdot [C_{t-1}, h_{t-1}, x_t] + b_f) \tag{5}$$

The output gate is shown in equation (6).

$$o_t = \sigma(W_o \cdot [C_{t-1}, h_{t-1}, x_t] + b_o) \tag{6}$$

As a variant structure of LSTM units, adding peephole connections is used to observe hidden information in neural networks and calculate the forgetting factor in the forgetting gate through short-term memory and temporal information. In fact, long-term memory does not participate in this computational process, and local short-term memory cannot provide global optimal decisions. The proposed peephole connection, by establishing a connection between forgetting factors and long-term memory, improves the decision-making participation ability of long-term memory under the global characteristics of neural networks.

Finally, introducing an attention mechanism to improve the LSTM network essentially simulates the resource allocation of human brain attention. At a specific moment, the human brain will focus its attention on the areas that need to be focused on, reducing attention to other areas in order to obtain more information that needs to be focused on and suppress other useless information. This can improve the situation where the LSTM network loses important information due to excessively long temporal data, and replace the random allocation of weights with probabilistic allocation of weights[6].Finally, the prediction results will be normalized to obtain the final prediction result.

E. Experimental analysis

The original load data in the example is shown in Figure 4, and the CEEMDAN load decomposition results are shown in Figure 5. As shown in the figure, CEEMDAN decomposes the load data into 6 IMF components and 1 residual component Res, effectively reducing the non-stationary nature of the original data.

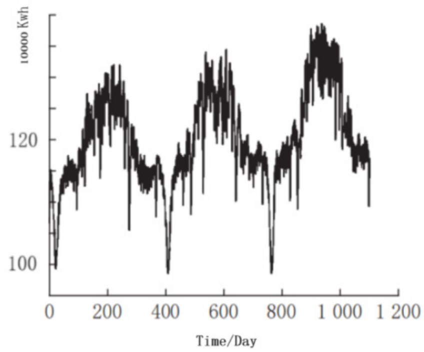

Fig.4 Original daily electricity consumption data.

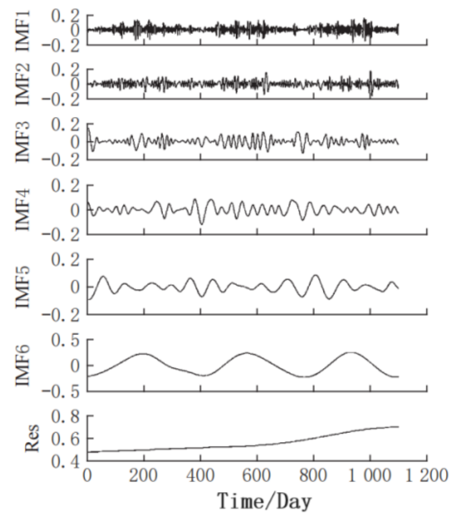

Fig.5 CEEMDAN Energy Decomposition Results

To further verify the accuracy of the CEEMDAN-LSTM-CNN network in predicting daily residential electricity consumption, comparative experiments were conducted with LSTM, CNN-LSTM, EMD-LSTM, and CEEMDAN-LSTM on the same training set and test set, respectively. To clearly observe the degree of fit between the predicted values of each model and the actual values, the predicted results of each model in the test set are presented in two parts. The predicted results of each model in the test set are shown in Figures 6 and 7, and the evaluation indicators and prediction accuracy of each model in the test set are shown in Table 1.

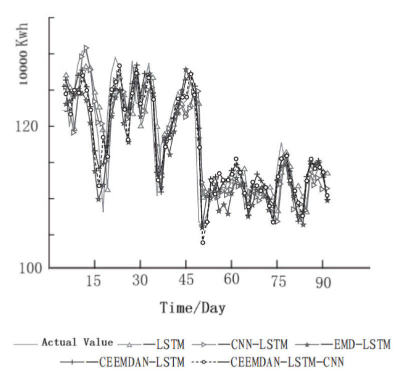

Fig.6 Comparison curve of predicted results in the first part of the test set.

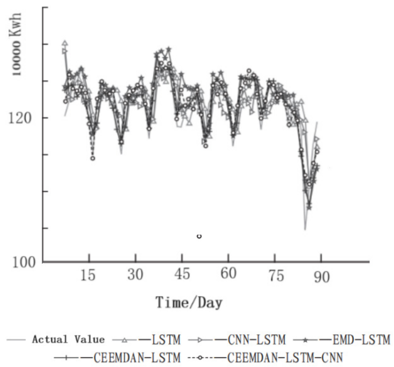

Fig.7 Comparison curve of prediction results in the second part of the test set.

Table 1 Evaluation indicators and accuracy of prediction results for each model

Model	Z_{MAE}/万 Kwh	Z_{RMSE}/万 Kwh	Z_{MAPE}/ %	Accuracy/%
LSTM	36.95	75.43	4.77	95.23
CNN-LSTM	28.98	66.21	4.37	95.63
EMD-LSTM	22.54	43.19	3.86	96.14
CEEMDAN-LSTM	18.31	28.43	3.18	96.82
CEEMDAN-LSTM-CNN	12.61	17.64	2.68	97.32

From figure 6 and figure 7, it can be seen that CEEMDAN-LSTM-CNN has a better grasp of the law of load changes, and its predicted curve is more in line with the actual load curve, resulting in better predictive performance. According to Table 1, the ZMAE, ZRMSE, and ZMAPE of CEEMDAN-LSTM-CNN have all decreased compared to the other four prediction methods, with a prediction accuracy of 97.32%.

III. Conclusions

Input an improved CNN layer, extract features from each decomposed subsequence, apply the ELU function as the CNN layer activation function, effectively solve the problem of mean shift, and accelerate the convergence speed. When the number of training iterations increases, overfitting or non convergence phenomena still do not occur. Add peephole connections in the LSTM layer to compensate for the potential loss of information by memory cells during output. The attention mechanism module is introduced to weight and reconstruct the predicted results of each output subsequence, ignoring irrelevant information and highlighting the role of important information, thereby improving the accuracy of the model.

By gradually improving the LSTM model and conducting simulation experiments during this process, and comparing the same set of test data with other highly applicable prediction methods in recent years, it is expected that the stationarity of time series with high complexity and strong nonlinearity will be improved during the prediction process, which will improve the accuracy of residential electricity consumption prediction.

Upon validation, the project has demonstrated strong practical guidance and has significantly enhanced the accuracy of power purchase agency forecasting. It provides a reference and replicable experience in excellent performance management for various levels of units within the system.

By the end of 2023, relying on the principles of excellent performance management, we aim to establish a scientifically efficient power purchase agency forecasting system. This will result in a significant improvement in the company's electricity forecasting indicators, a notable reduction in factors affecting forecasting accuracy across various units, and an annual prediction error rate for residential and agricultural power supply not exceeding 4.8%, an error rate for power purchase agency forecasting not exceeding 5.1%, and a comprehensive error rate not exceeding 5%. This will position us at the forefront of the province.

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